

Simple Tester for 74xx04 and 74xx14 ICs

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The integrated circuits (ICs) 74xx04 and 74xx14 are very frequently used and reused during circuit design. Here xx stands for HC, HCT, AC and LS, etc. Sometimes, the internal inverters of these ICs get damaged. So, it is very important to

test these ICs before initiating any experiment and sometimes during the experiments.

Test Points for Gate N1

Test points	Input	Output
J1 closed but J7 open	High at TP2	Low at TP3+LED1 on
J1 open but J7 closed	Low at TP2	High at TP3+LED7 on
J1, J7 and J13 open	Oscillators enabled	At TP3—frequencies vary from 1kHz to 182kHz depending on the position of preset VR1
J13 closed and J1+J7 open	Oscillators enabled	At TP3—frequency is around 0.5Hz. Both LEDs blink

Presented here is a simple circuit which can test these ICs statically and dynamically.

Circuit and working

Fig. 1 shows the circuit diagram of a simple tester for 74xx04 and 74xx14 ICs. The circuit is built around hex inverter IC 74HC04 (IC1) and a few other components. You can also test a 74HC14 hex Schmitt trigger with this circuit.

Static testing. The jumpers J1 through J12 are used during static testing of the inverter IC. When J1 through J6 are closed, J7 through J12 should be kept open, and vice versa. When jumpers J1 through J6 are closed, the input of the corresponding inverter is

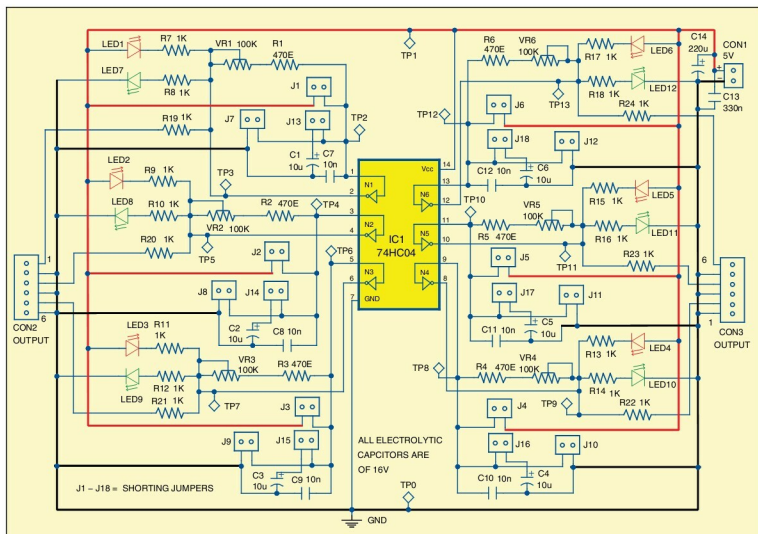


Fig. 1: Circuit diagram of the simple tester for 74xx04 and 74xx14 ICs